

Coiled Tubing Introduction

- Surface equipment
- BHA assemblies
- Some problems and observations

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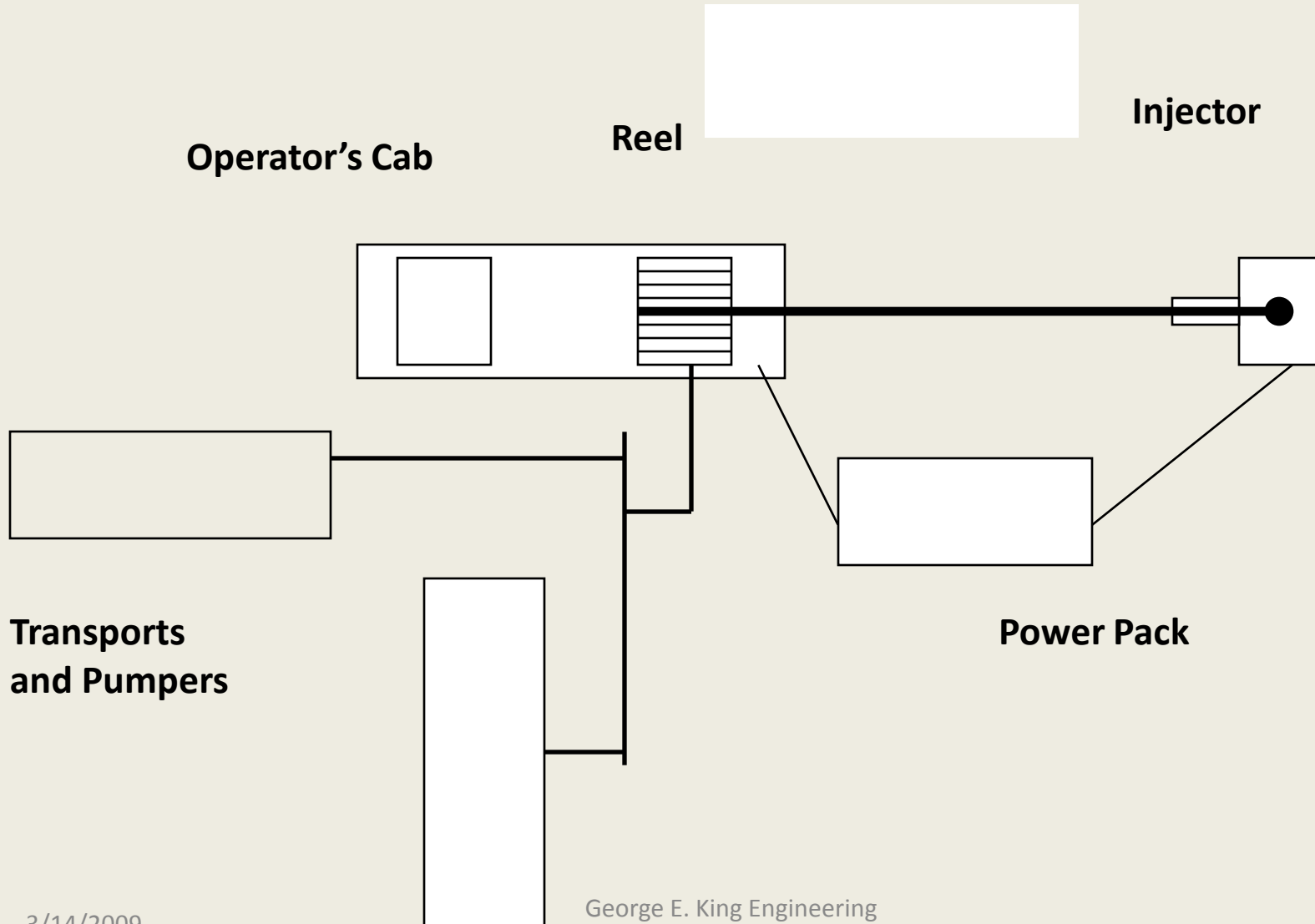


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	World Wide CT Records
Largest CT in Use	3-1/2"
Max Depth	24,000'
Max Horizontal	17,000' (80 deg)-Wytch Bl
Longest BHA	1500' - perf guns - UK
Longest Strings	23,000' of 2-3/8"
	28,000' of 1-1/2
Max Wellhead Pressure	9800 psi
Max Deployment Press	4500 psi
Max BH Temp	700F - Mex, 780F Japan
Max Acid at Temp	28% at 280F in Dubai
CT in H2S	75% in Greece
	98%/300F- Gulf of Mexico
	(string used one time)
CT in CO2	15%- (string ruined)

Typical CT Layout



CT Reel Configurations

- CT Reels available in number of configurations
 - Truck mounted (fixed) - permanently fixed to the truck chassis
 - Truck mounted (skid) - may be changed out
 - Skid mounted - for offshore operations
 - Trailer mounted - for large capacity (length) or heavy weight strings
 - CT logging reel - fitted with electrical swivel/collector
 - Special application reel - typically for completion applications
- Local conditions/nature of CT operations determine type of reel required



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Coiled Tubing Rig up in
Alaska. Two wells on the
same pad are being.





CT Equipment Basics

- Components
 - Coil
 - Power pack
 - Component Controls
 - Reel
 - Injector (with guidearch)
 - Well Control Components

Coil Types and Availability

- CT diameters
- CT wall thickness
- CT tapered strings
- CT strength ranges
- CT metal choices
- Composite CT

CT Diameters

- 1/2, 3/4 and 1" early diameters
- 1-1/4" and 1-1/2" workhorses (1-3/4)
- 1-3/4 to 2-3/8" larger workstrings (3-1/2)
- 3-1/2 and 4-1/2" (and larger) - flowlines

CT production for For 1996

- 1" 1.52 MM ft
- 1.25" 14.2 MM ft
- 1.5" 6.2 MM ft
- 1.75" 3.3 MM ft
- 2" 1.22 MM ft
- 2-3/8" 0.56 MM ft
- 2-7/8" 0.43 MM ft
- 3-1/2 0.45 MM ft
- 4&4.5" 0.125 MM ft

Tapered Strings

- Common is I.D. taper - weld area taper
- Uncommon is O.D. taper - deeper wells, some hydraulic problems

**Butt Weld – early
string joiner and
common repair weld.**

**Junction weld
in flat strip**



The bias weld offsets the weld ends on the strip as it is formed into a tube – much stronger.

**Bias weld
in flat strip**

Field Welds

- Always a Butt weld.
- Can decrease tensile strength of CT by 50% or more.
- Quality of field welds varies enormously
- Avoid field welds when you can.
 - Upper part of tubing is worst place for a butt weld.

Material Strengths

- 55,000 psi (limited)
- 70,000 to 80,000 psi (still in use)
- 100,000+ psi (now very common)

Trade-Offs

- strength vs cost
 - ductility/weakness vs strength/cost
 - corrosion possibilities
 - strength vs service life
-
- In general, high strength CT lasts longer and is more fatigue resistant. But, what about corrosion? Hardness matters. Stress Corrosion cracking and embrittlement.

Analysis of Coiled Tubing

	Mod A606	Mod A607
Carbon	0.08-0.15	0.08-0.17
Manganese	0.60-0.90	0.60-0.90
Phosphorus	0.030 max	0.025 max
Sulfur	0.005 max	0.005 max
Silicon	0.30-0.50	0.30-0.45
Chromium	0.45-0.70	0.40-0.60
Nickel	0.25 max	0.10 max
Copper	0.40 max	0.40 max
Molibidum	0.21 max	0.08-0.15
Cb-V	0.02-0.04	

New CT Properties

Yield Strength	HRB	HRC	Min. Elong.
(psi)			
50,000		22	
70,000	85-94		30%
80,000	90-98		28%
90,000	94	22	25%
100,000		20-25	22% est.
110,000		22-28	18% est
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Composites

- Limited Use
- Composites lighter/more flexible than steel.
- More costly than steel
- Limitations:
 - Creep under load
 - Temperature
 - Downhole Buckling resistance

Surface Equipment

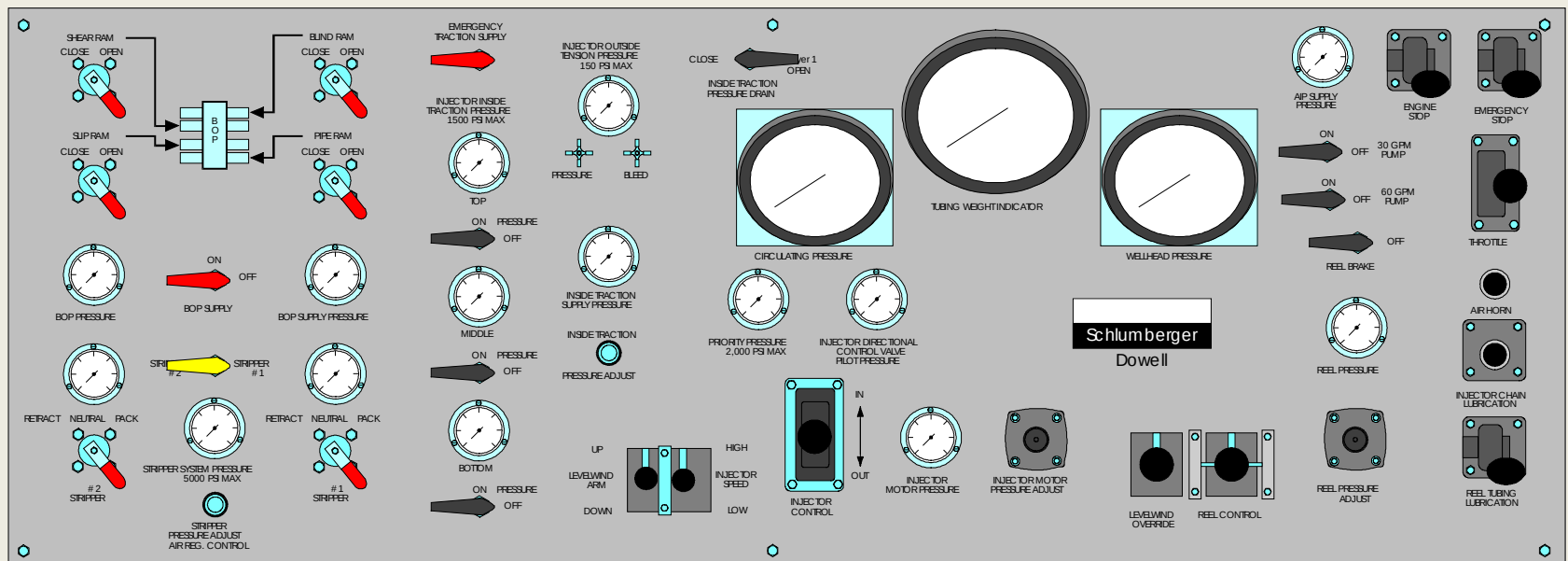
- Power Pack
- Reel
- Controls
- Injector
- Seals

Power pack – usually diesel driven hydraulic pumps.

Requirements set by equipment in use.



Typical Layout of Control Cabin Controls and Instruments (1)



Looks more complex, but main data is from load, wellhead pressure and annular pressure.

Main controls are injector head and coil spooling.



BDN 1.75

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PRIORITY
PRESSURE

SYSTEM AIR
PRESSURE



CLOSE



EMERGENCY
KILL

INJECTOR INSIDE
TRACTION PRESSURE



TOP
ON

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Principal Gauges

—Wellhead Pressure Gauge

- displays wellhead pressure at the BOP pressure port

—Circulating Pressure Gauge

- displays pressure at the reel-manifold pressure sensor

—Weight Indicator

- displays weight exerted by the tubing on the injector head

CTU Monitor

Zoom in

Pressures

Zoom out

Zoom in

String Weight

Zoom out



Wellhead PSI: 67
Tubing PSI: 726
Annulus PSI: 202
Pump PSI: 689

Rate 1	Rate 2	MM Rate	Density
0.8	2.8	0.8	8.40
Total 1: 334.3	Total 2: 11439.1	MM Total: 86.8	Temp: 91.8

Speed Ft./Min.

178.8

ROP Ft./hr.

10723

Depth / Ft.

2772.8

CTD / ft.

12958.5

C-Pump 1

56

C-Pump 2

280

Weight / Lbs.

5432

WOB

0

Tag

Auto Lube

On

Off

End



Reset Inj.

Shortcut to Driller

0.465 1 Point

60 25 50 75 100

393319

Start

Setup and Control

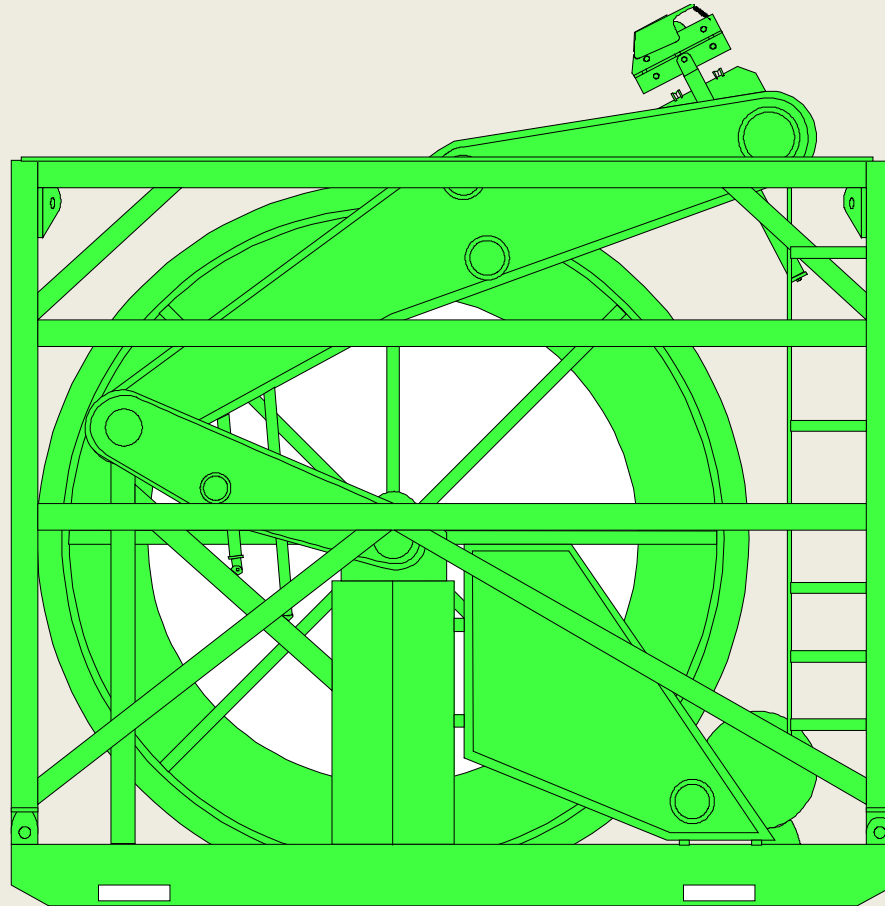
CTU Monitor

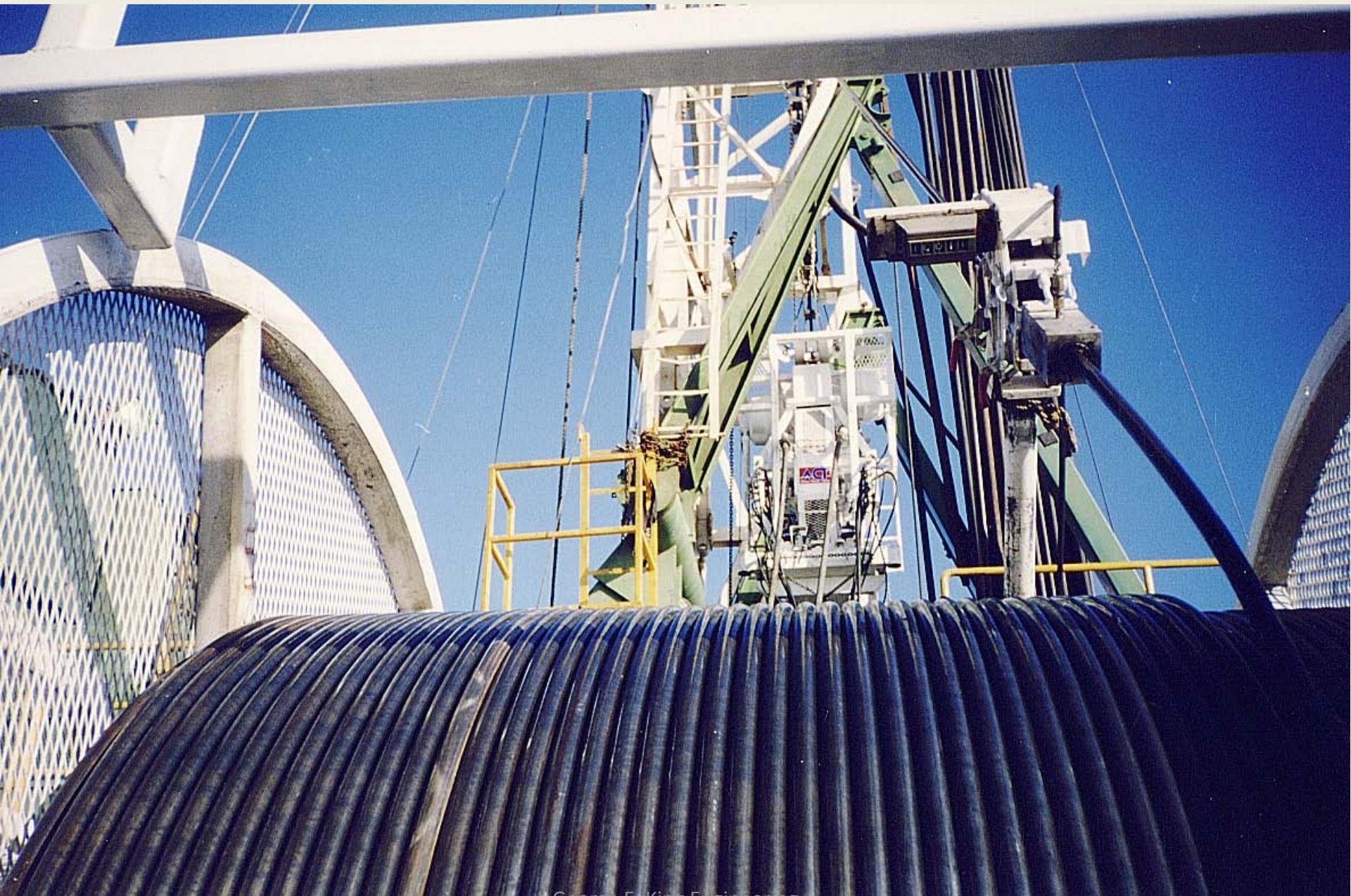
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Running Speeds

- First time into a well - 40 to 70 ft/min
- Normal run in operations - 50 to 100 ft/min
- Spool-out? – depends on the well, the BHA, coil equipment, friction and operator.
- Runaway - experienced 800 to 900 ft/min.

Typical CT Reel - Side View

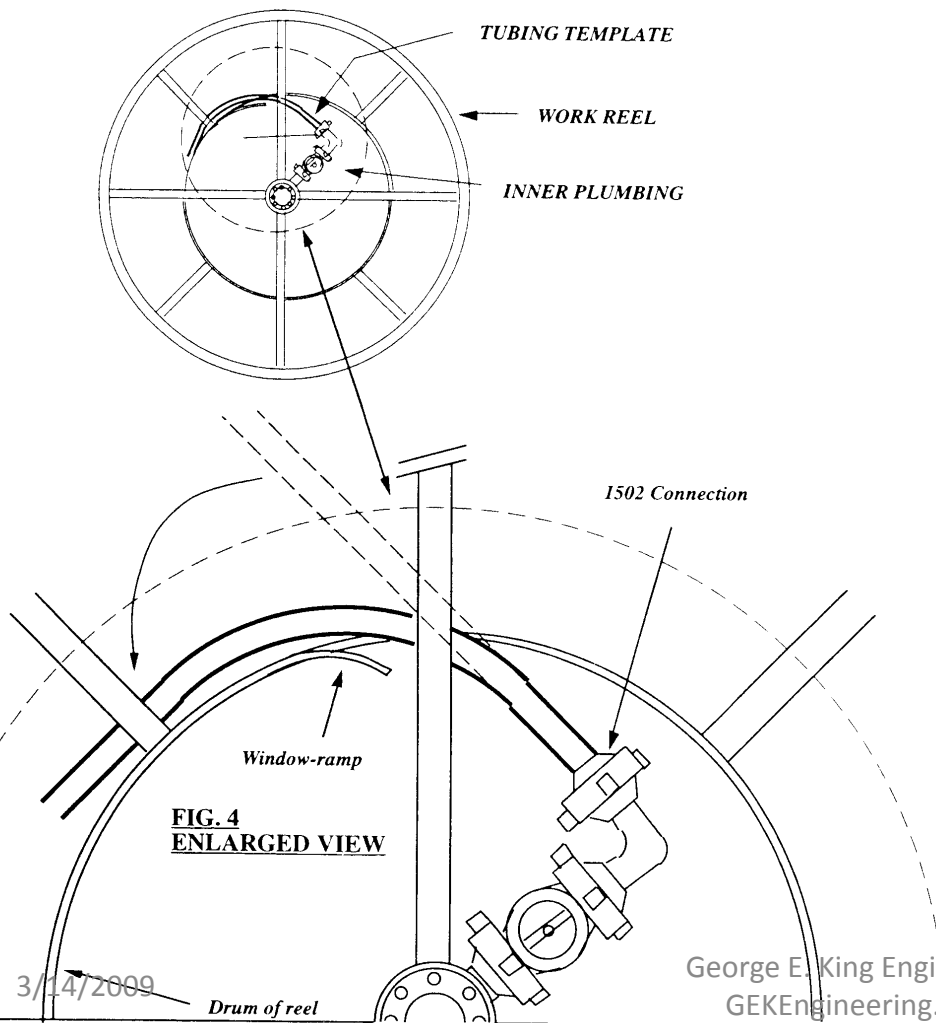


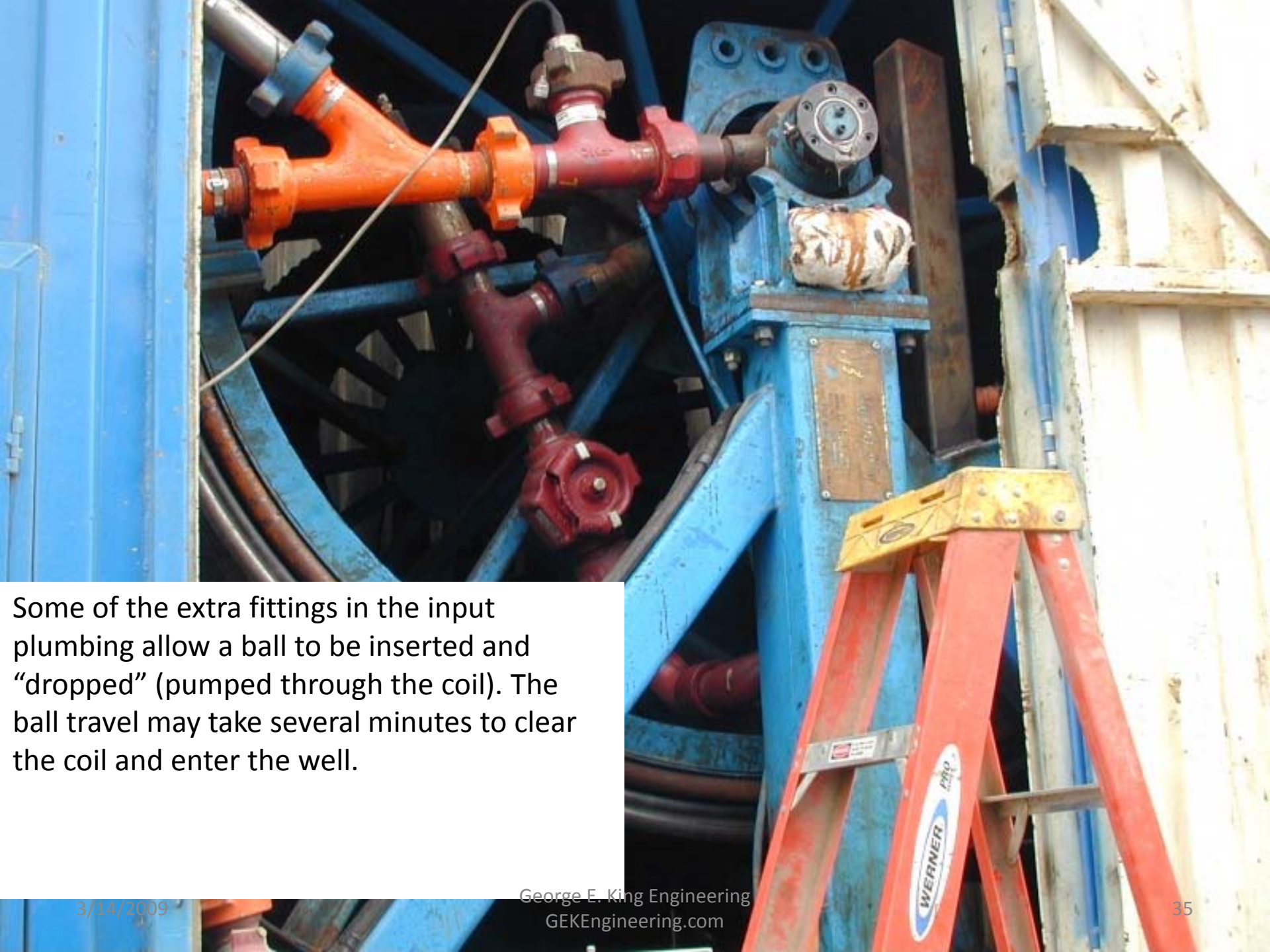


What's Different Than Jointed Pipe?

All fluids injected from the surface enter the CT at the bed-wrap end. All fluid has to travel through all of the coil.

That influences friction since friction is near constant w/ CT whether you are injecting fluid at 1 foot into the well or 10,000 ft into the well.



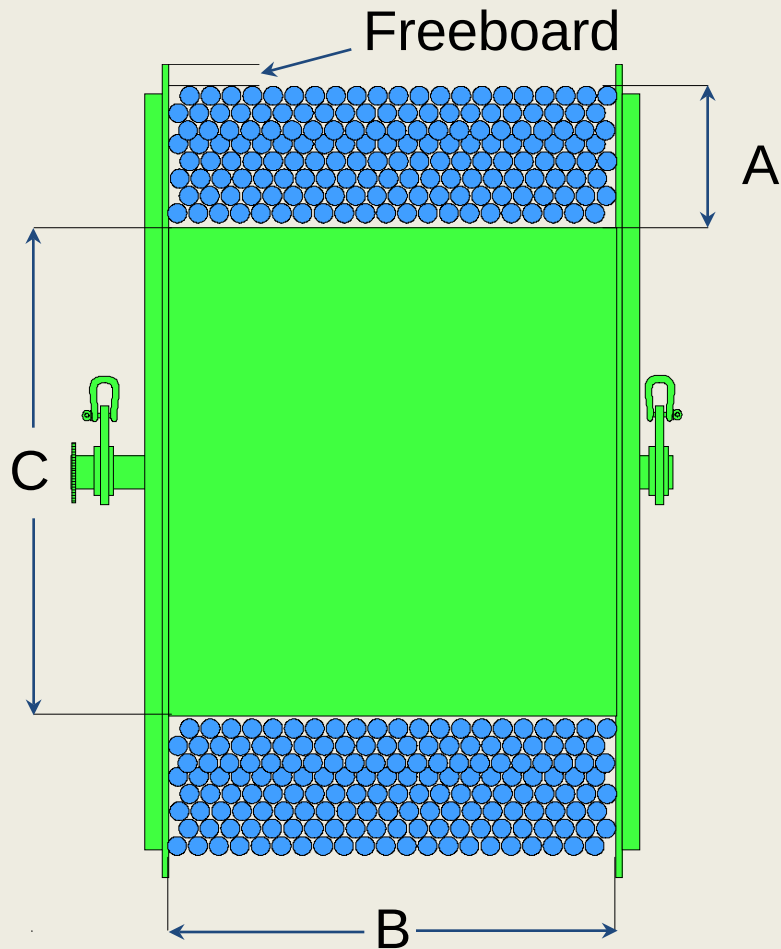


Some of the extra fittings in the input plumbing allow a ball to be inserted and “dropped” (pumped through the coil). The ball travel may take several minutes to clear the coil and enter the well.

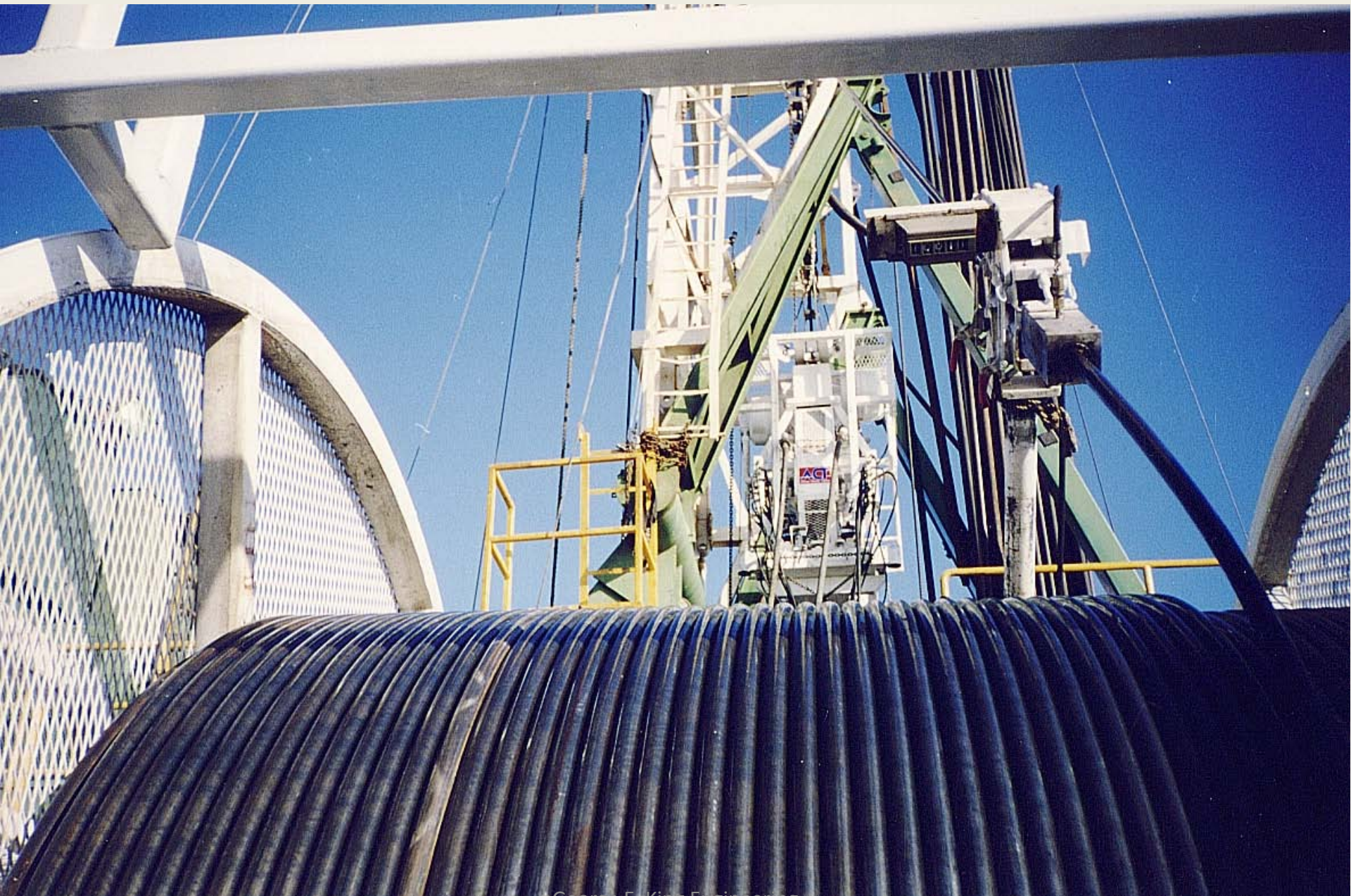
A flow control manifold. This one allow reverse circulation through coiled tubing.



Reel Drum Capacity



Freeboard is normally not much of an issue, but can become very important if the CT is not spooled tightly and the drum is nearly full for a deep well CT job.





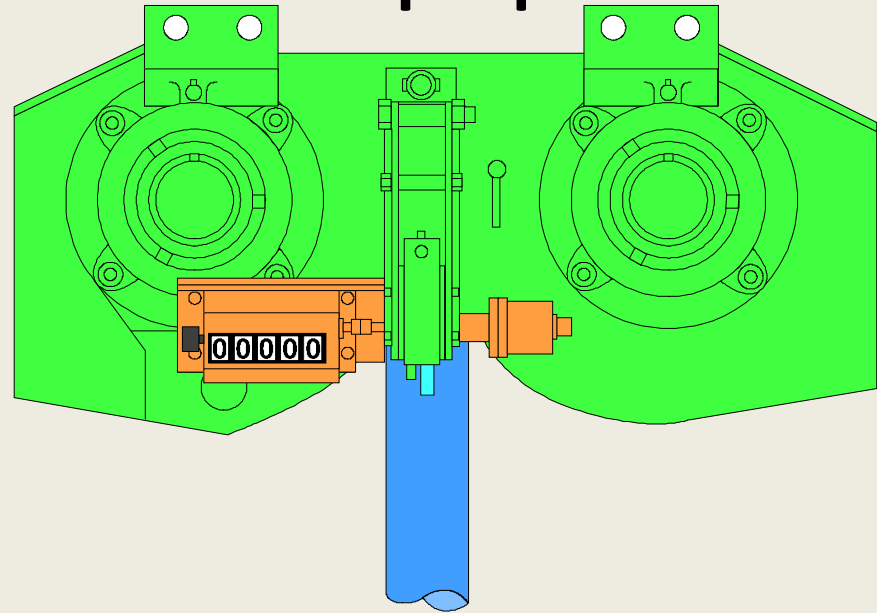
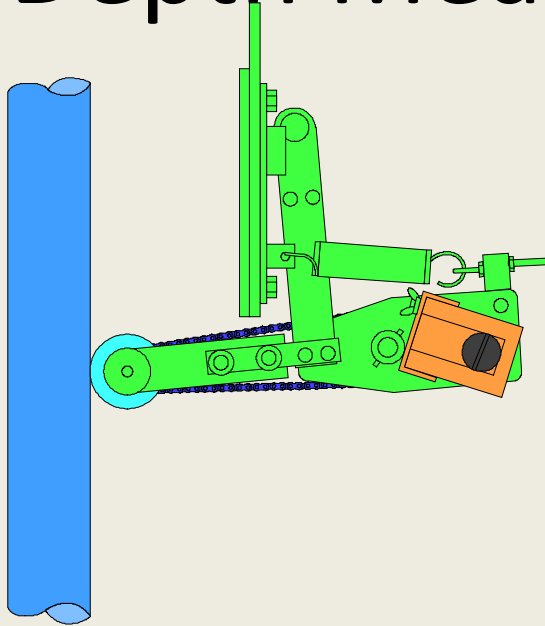
Selected Reel Diameter Information

Reel Information for Various Sizes of CT				
CT Diam.	Reel Hub Diam.	Guide Arch	(Reel Width)	Approx. Capacity
(in)	(in)	(in)	(in)	(ft)
0.75	48	48		
1	60	48-54		
1.25	72	48-72	117	17500
1.5	84	(48)-72	128	15000
1.75	96	72-96	148	18000
2	96	72-96		
2.375	108	90-120		
2.875	108	90-120		
3.5	120	90-120		

The level-wind (like a bait-casting fishing reel) helps feed the coil onto the reel.



Depth Measurement Equipment

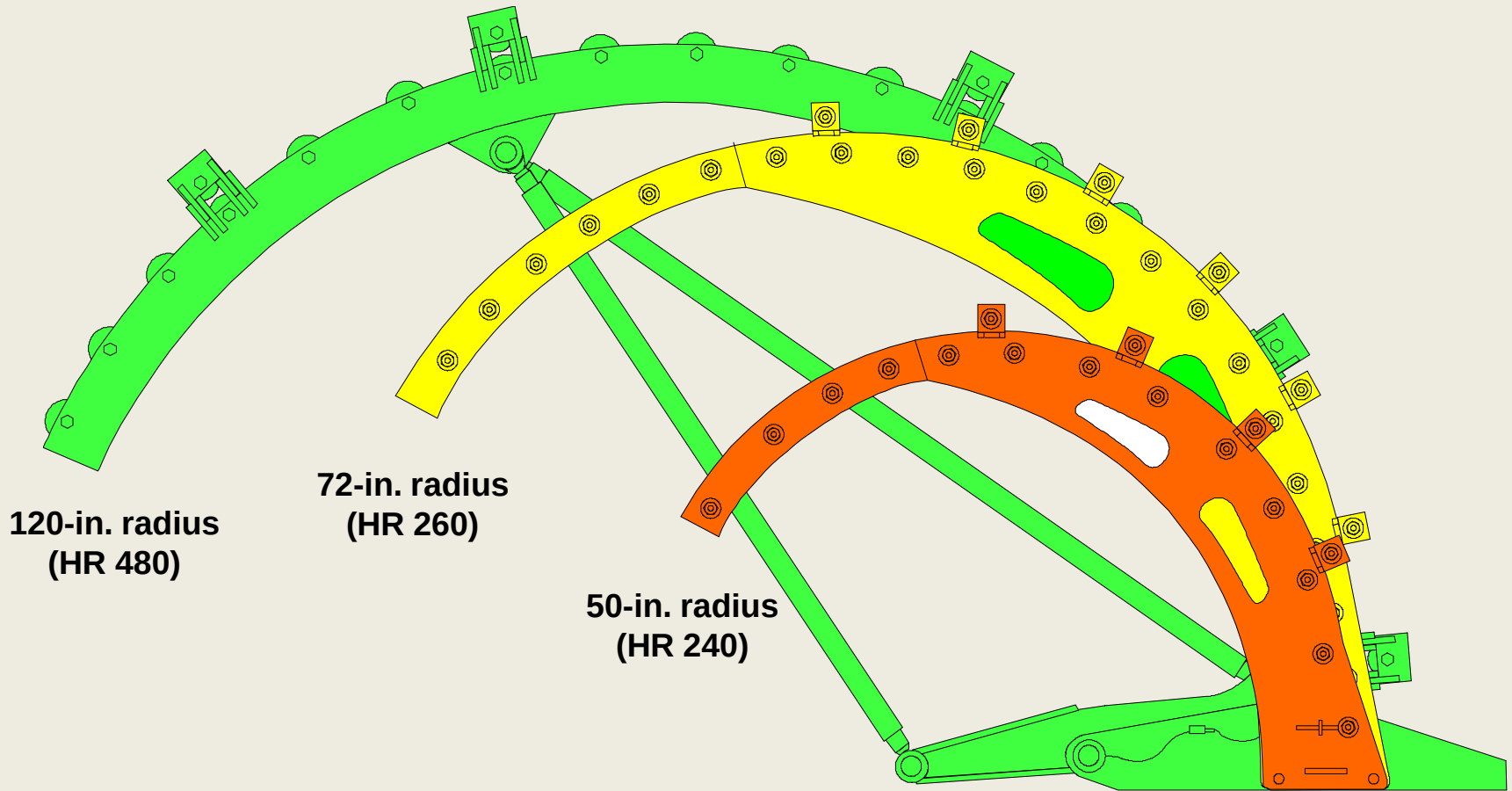


- Depth measuring equipment -Electronic or mechanical
 - Frequently mounted on injector head
- Depth information commonly acquired by two methods:
 - Friction-wheel counter between injector chains and stripper
 - Encoder assembly on injector-head chain drive shaft

Injector Guide Arch

- Guide arch
 - Turn tubing through angle between wellhead and CT reel
 - CT supported by rollers at ± 10 -in. intervals around gooseneck
 - guide arch radius significantly affects fatigue in CT string

Comparison of Guide Arch Sizes



Recommended Guide Arch Radii

Coiled Tubing OD (in.)	Typical Reel Core Radius (in.)	Typical Guide Arch Radius (in.)
0.750	24	48
1.000	20-30	48-54
1.250	25-36	48-72
1.500	30-40	48-72
1.750	35-48	72-96
2.000	40-48	72-96
2.375	48-54	90-120
2.875	54-58	90-120
3.500	65-70	96-120

CT Injectors

- Injector
- Gripper Design and Effect on the Tube
- Power Requirements
- Load Cells
- Wear and Failure Points
- Control vs “micro” control

Injector Head Components

- Principal components of injector head:

- Drive and brake system
- Chain assembly
- Traction and tension system
- Guide-arch assembly

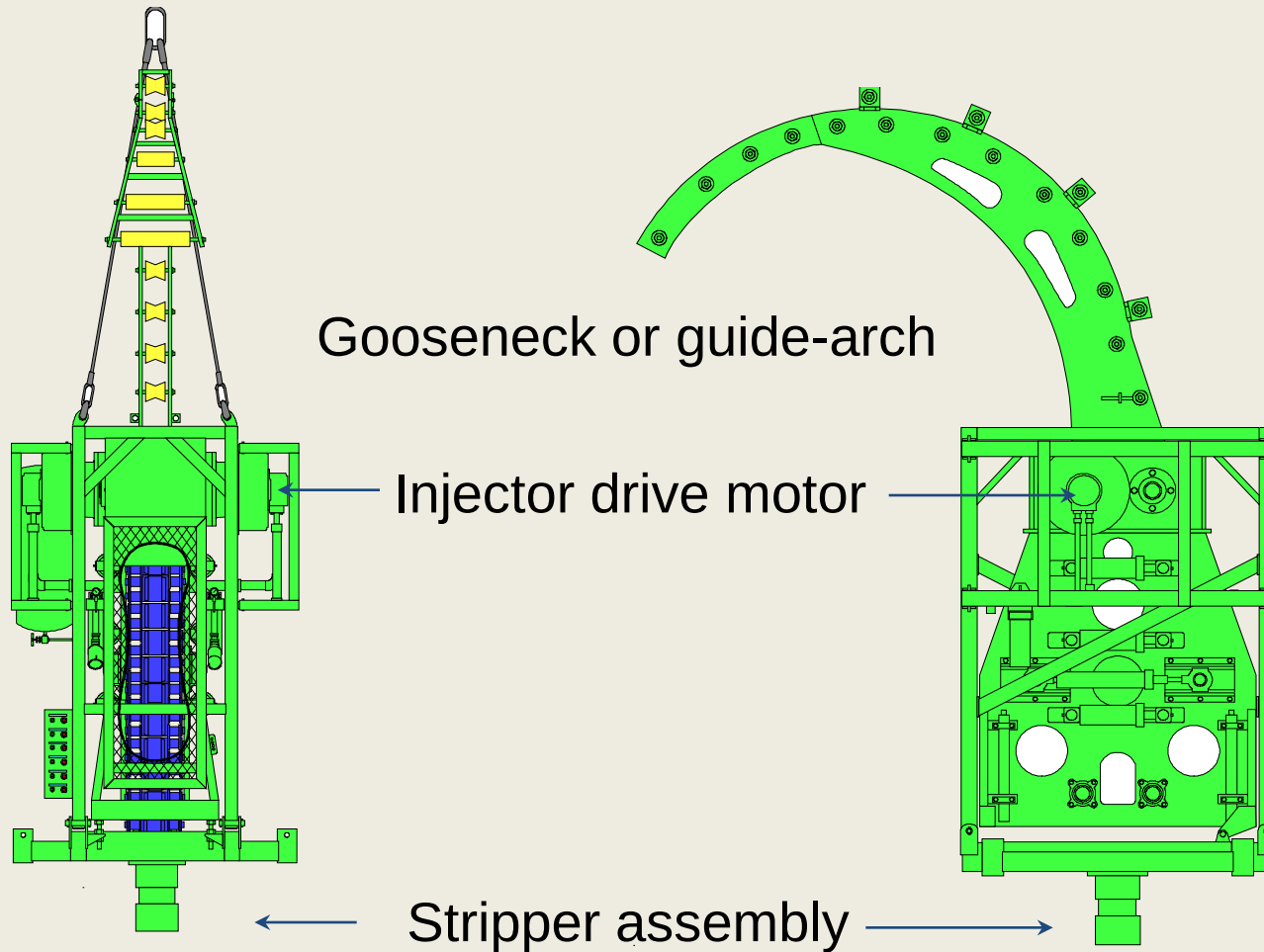
- Secondary or support systems:

- Weight indicator
- Depth sensor mounts
- Stripper mount

Injector Head Drive Systems

- Injector heads hydraulically driven
 - Two or four hydraulic motors
 - Motors connected/synchronized through gear system
 - Drive directed to chain drive sprockets via drive shafts
 - Rotation/speed of motors controlled by valve on power pack
 - Hydraulic system pressure/rate controlled from CTU control console
 - Pressure relief/crossover relief valves
 - protect tubing/hydraulic components

Typical Injector Head





Injector Head Brake Systems

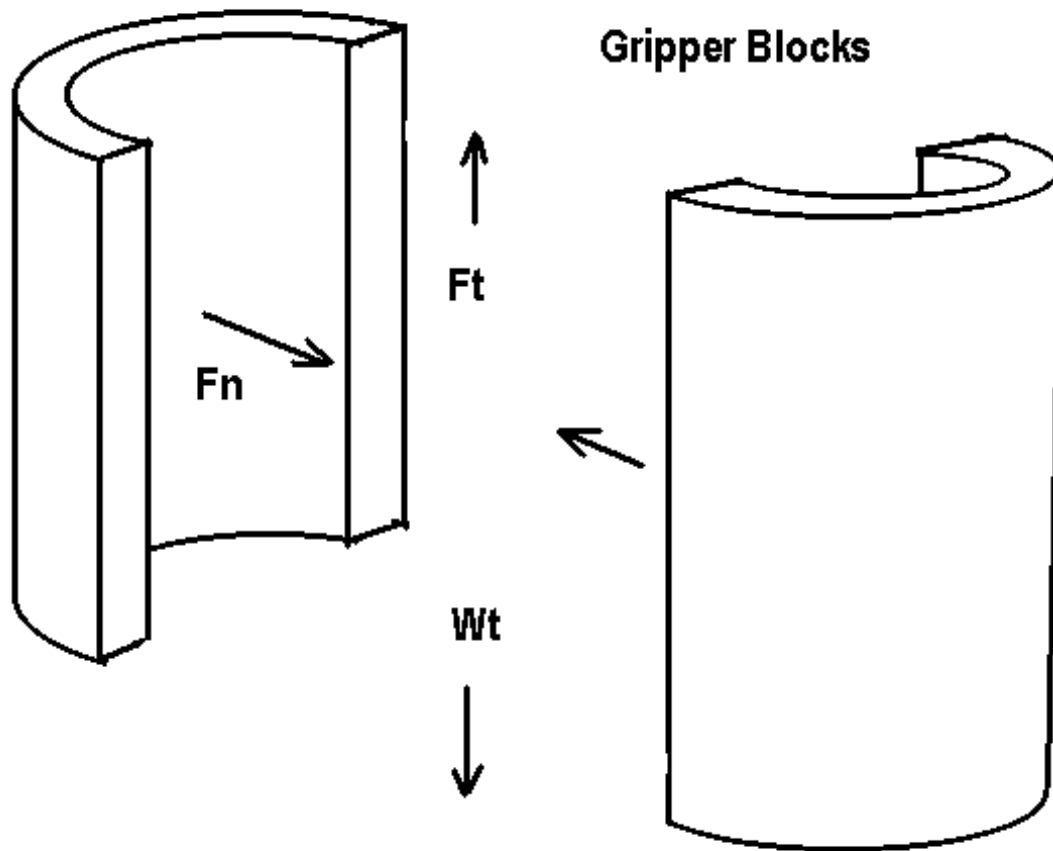
- Brake hydraulically controlled
 - Hydraulic pressure required to release brake
 - fail-safe in operation
 - Application of brake is automatic
 - controlled by drive system hydraulic pressure
 - applied when hydraulic pressure below preset value
- Some hydraulic motors have high/low gear option
 - Selected remotely from the CTU the control console
 - Allows injector head to operate more efficiently

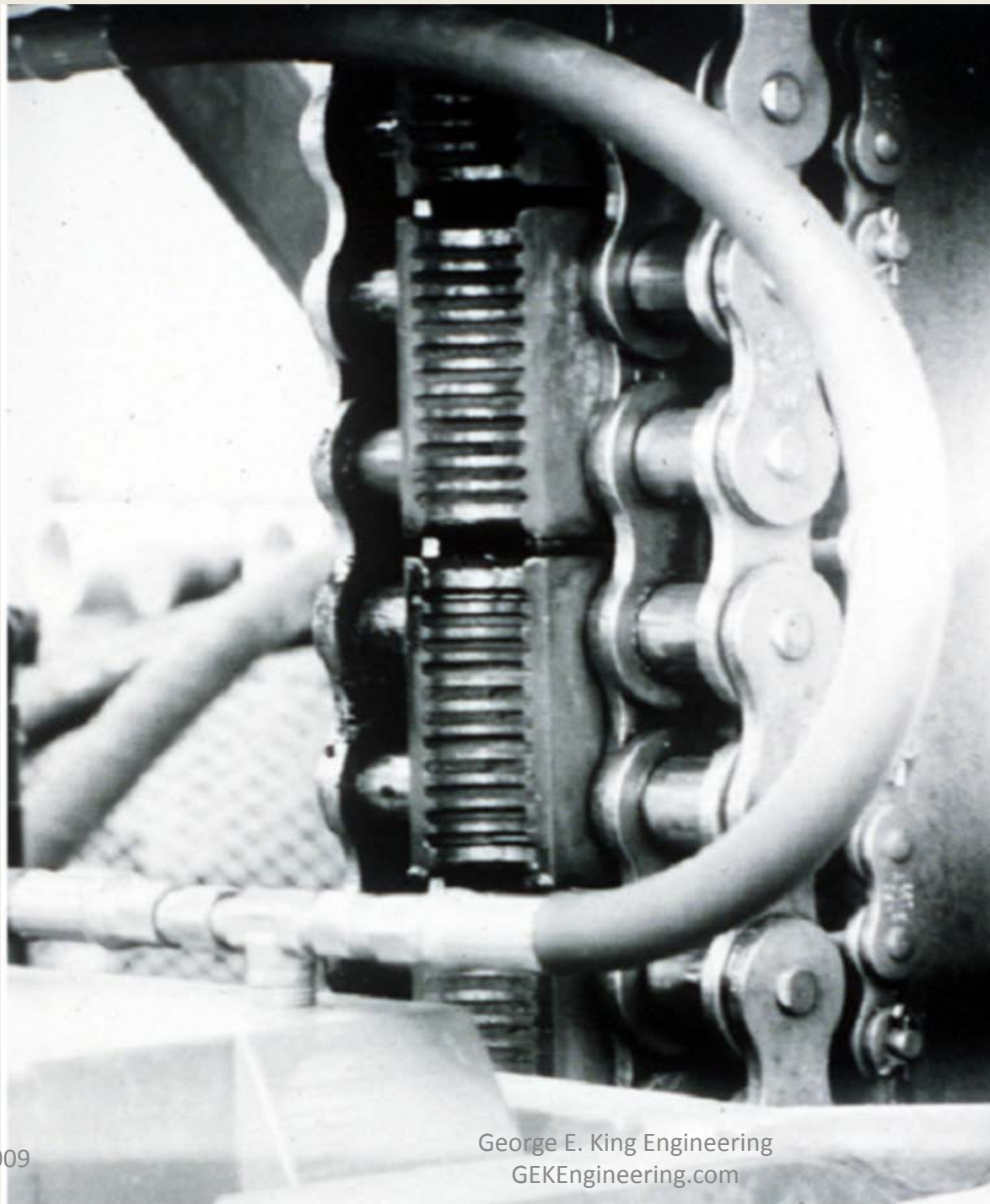
Injector Head Chain Assembly

- Most injector heads have two sets of opposing endless chains
 - Series of gripper blocks mounted on each chain
 - Gripping profile of each block suits a specific tubing size
- Entire CT string load held by face of the gripper block or insert
 - Often achieved under significant force
 - Selection/operation/maintenance of chain components should minimize risk of damage to CT string

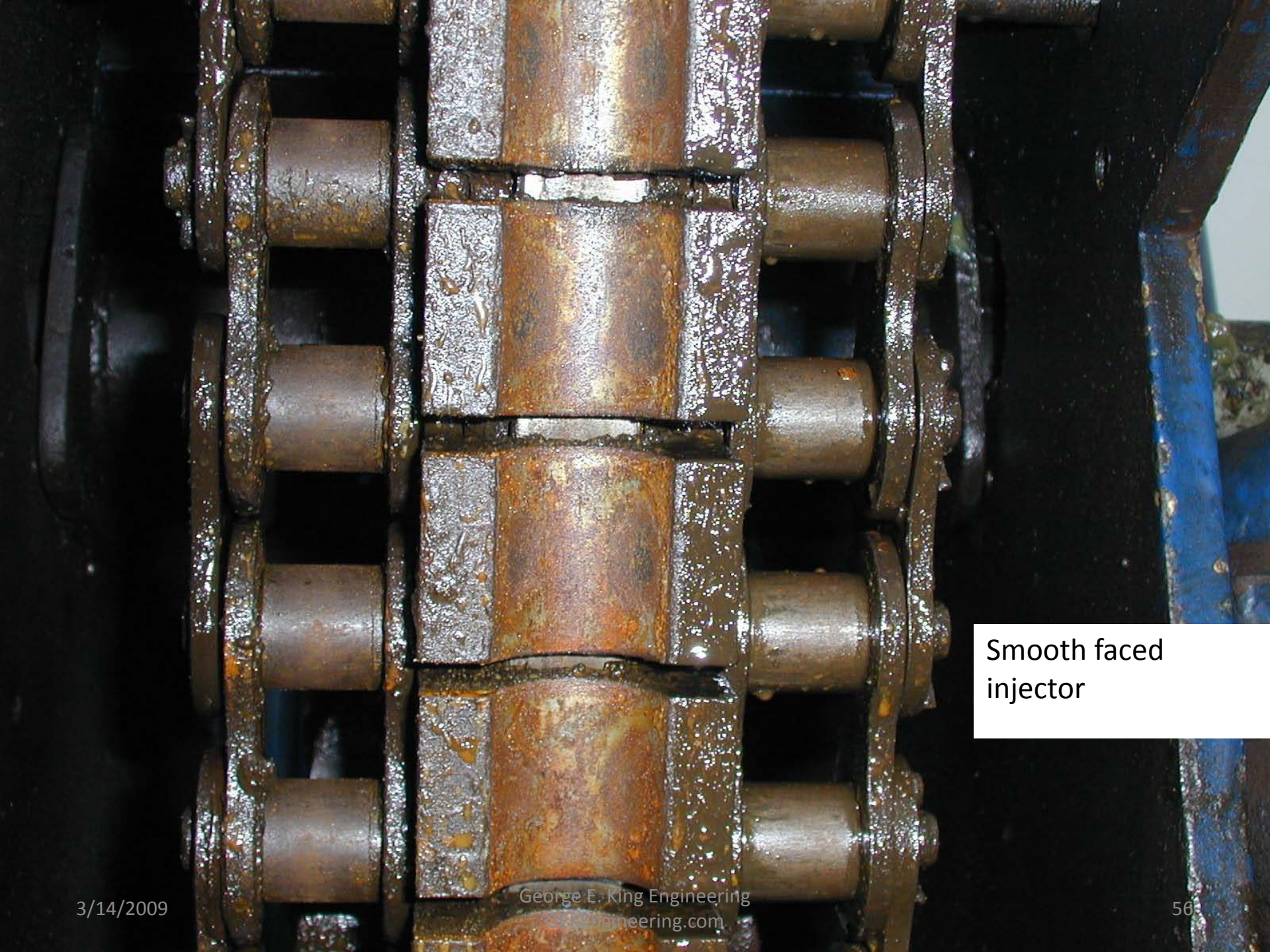
Injector Head Chain Gripper Inserts

- Removable gripper inserts on most injector heads
 - Run range of tubing sizes without removing/replacing entire chain assembly
 - Reduce time/effort required to reconfigure injector head
 - when running a different size of CT string
 - when running a tapered CT string
 - Reduce cost of replacement when worn



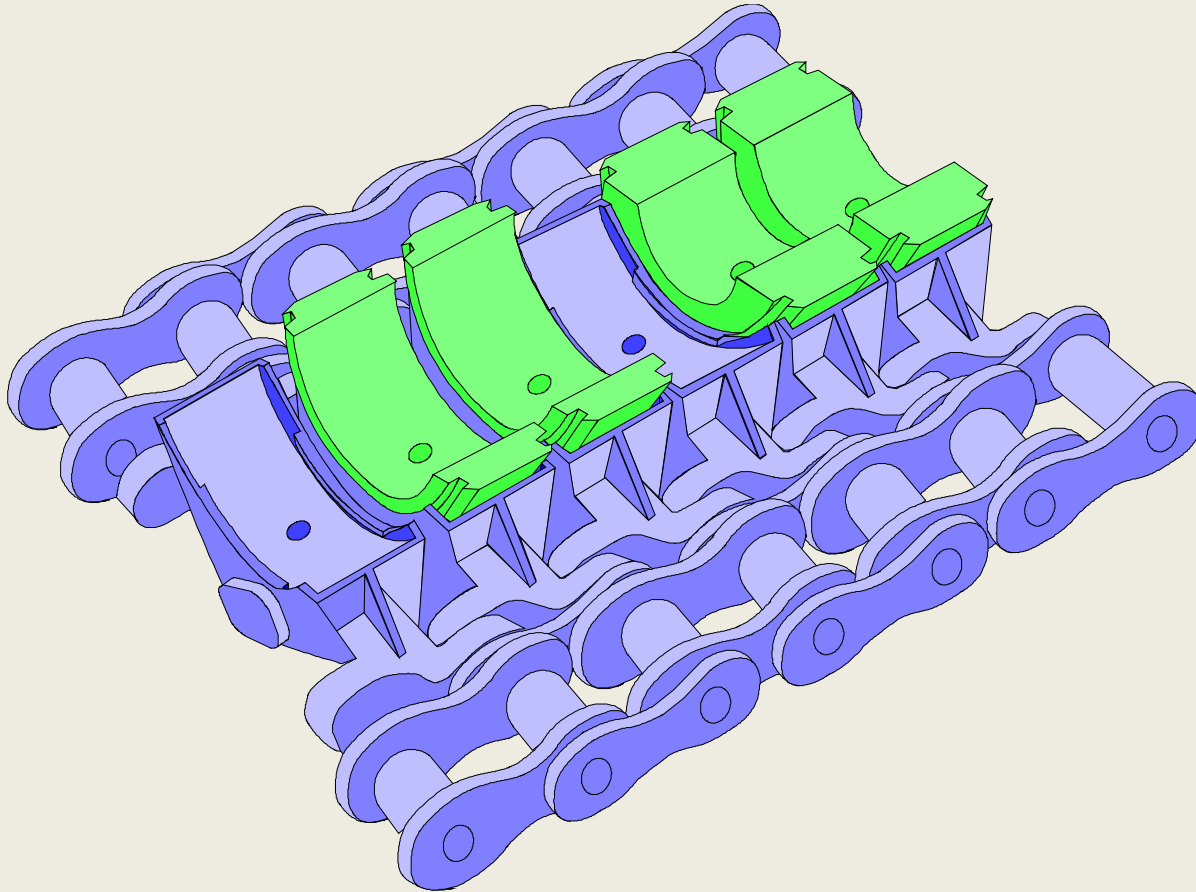


Ribbed injector block.



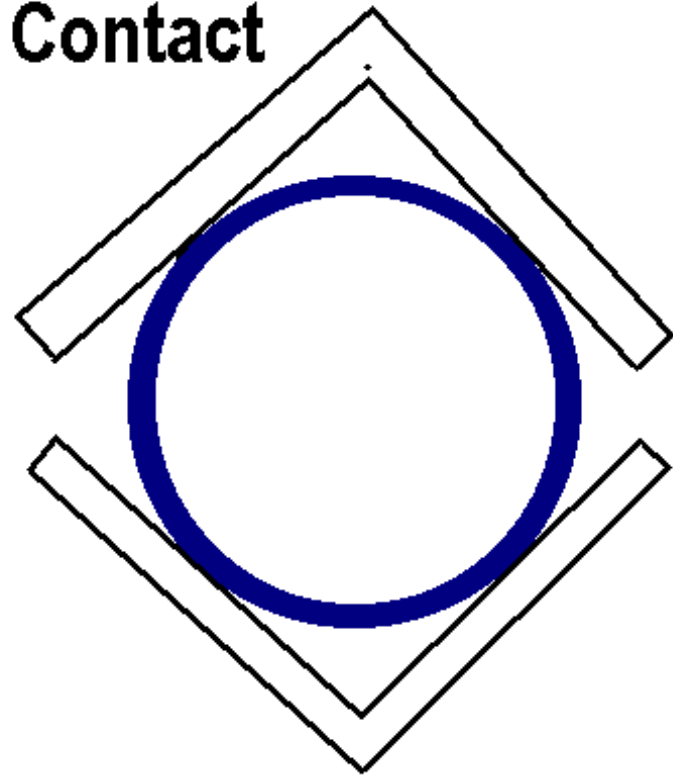
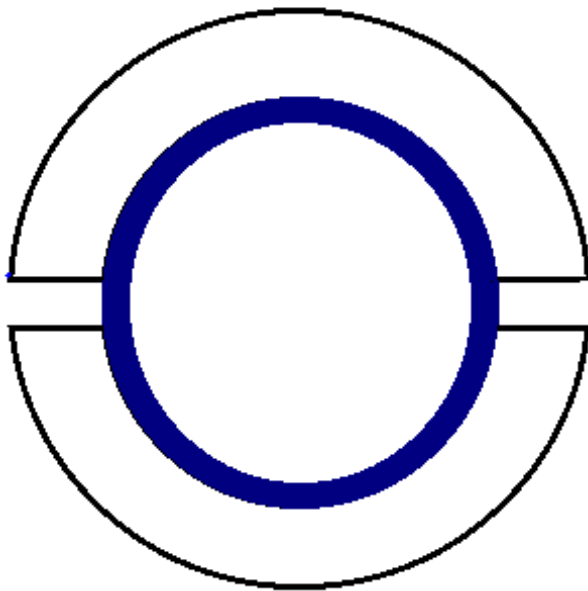
Smooth faced
injector

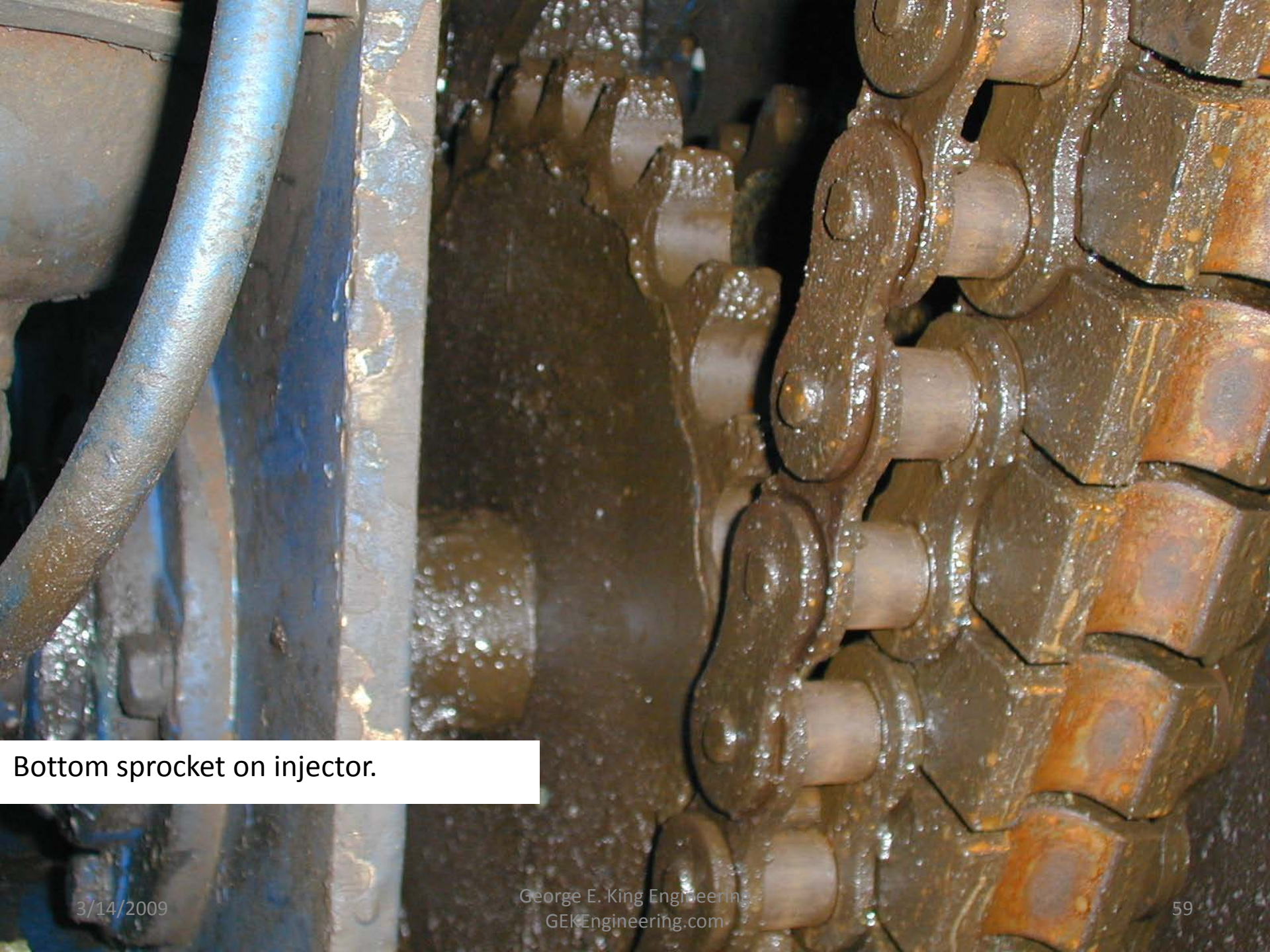
SS800 Chain Assembly



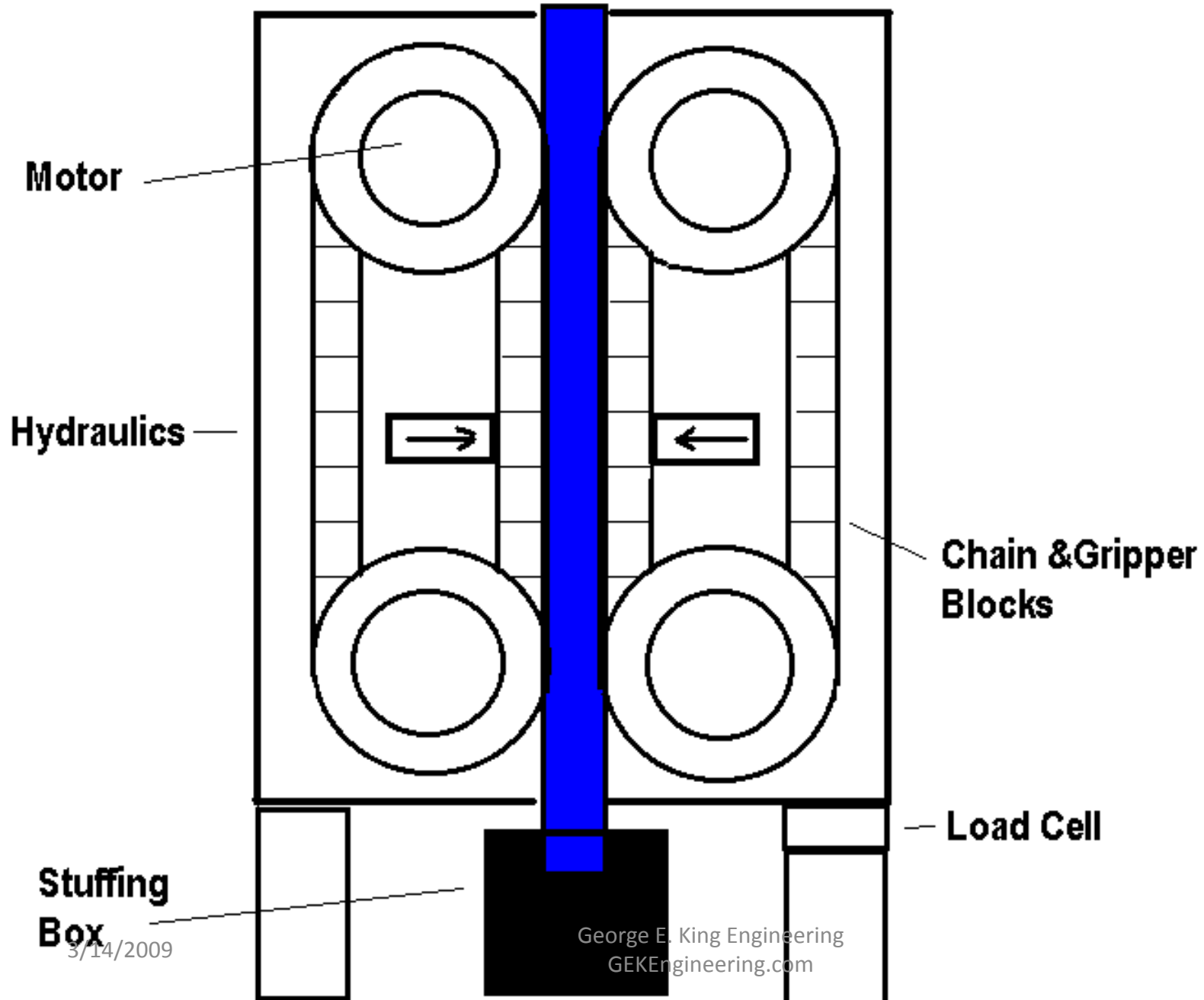
Gripper Block Type

Utility vs Contact





Bottom sprocket on injector.

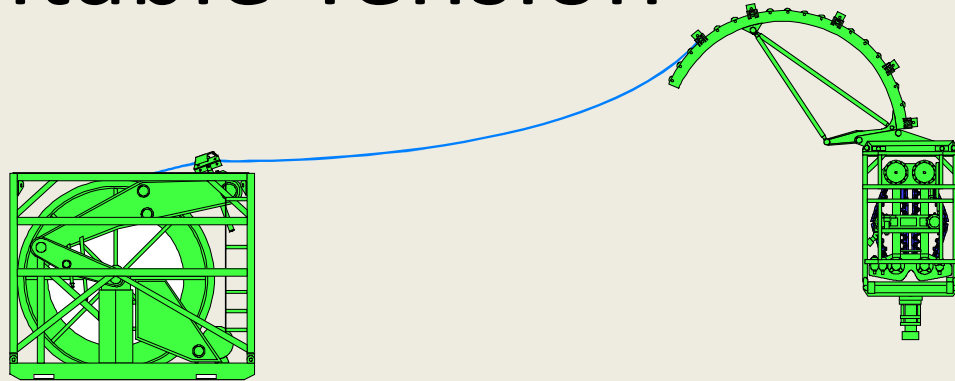


Traction and Tension Systems

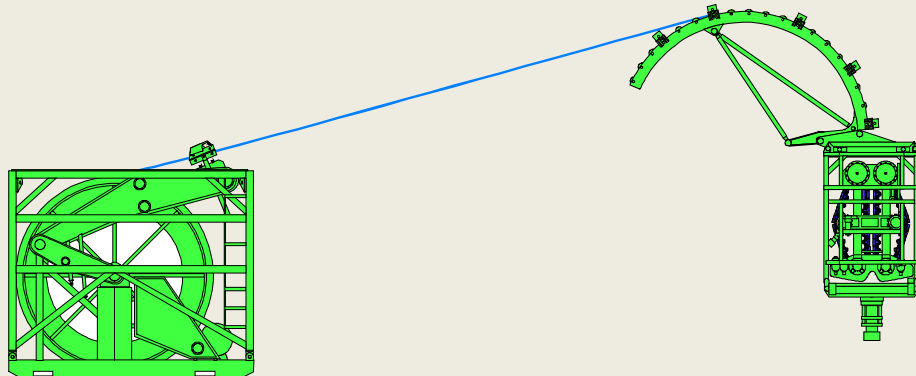
- Injector head traction or inside chain tensioner system
 - Also known as skate system
 - Provides force to securely grip tubing
 - Force applied through 3 separate sets of hydraulic cylinders
 - reduces risk of major operating failure if component fails
- Outside chain tensioner system
 - Ensures adequate tension in chain section outside vertical drive plane
 - Tension provided by hydraulic rams

Backpressure Adjusted to Provide Suitable Tension

Incorrect



Correct







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Coiled Tubing
Deployment

BHA

Deployment BOP

179.4 mm, 35 MPa

Flow Line,
101.6 mm

ESD
(Tester)

Coiled Tubing BOP

179.4 mm, 35 MPa

Drilling Spool,

179.4 mm, 35 MPa
Kill Line
, 50.8 mm

Gate Valve,
179.4 mm

Locating
Ram
Upper
Pipe Ram

Lower Pipe
/Slip Ram

Flow Spool

Annular
Preventer

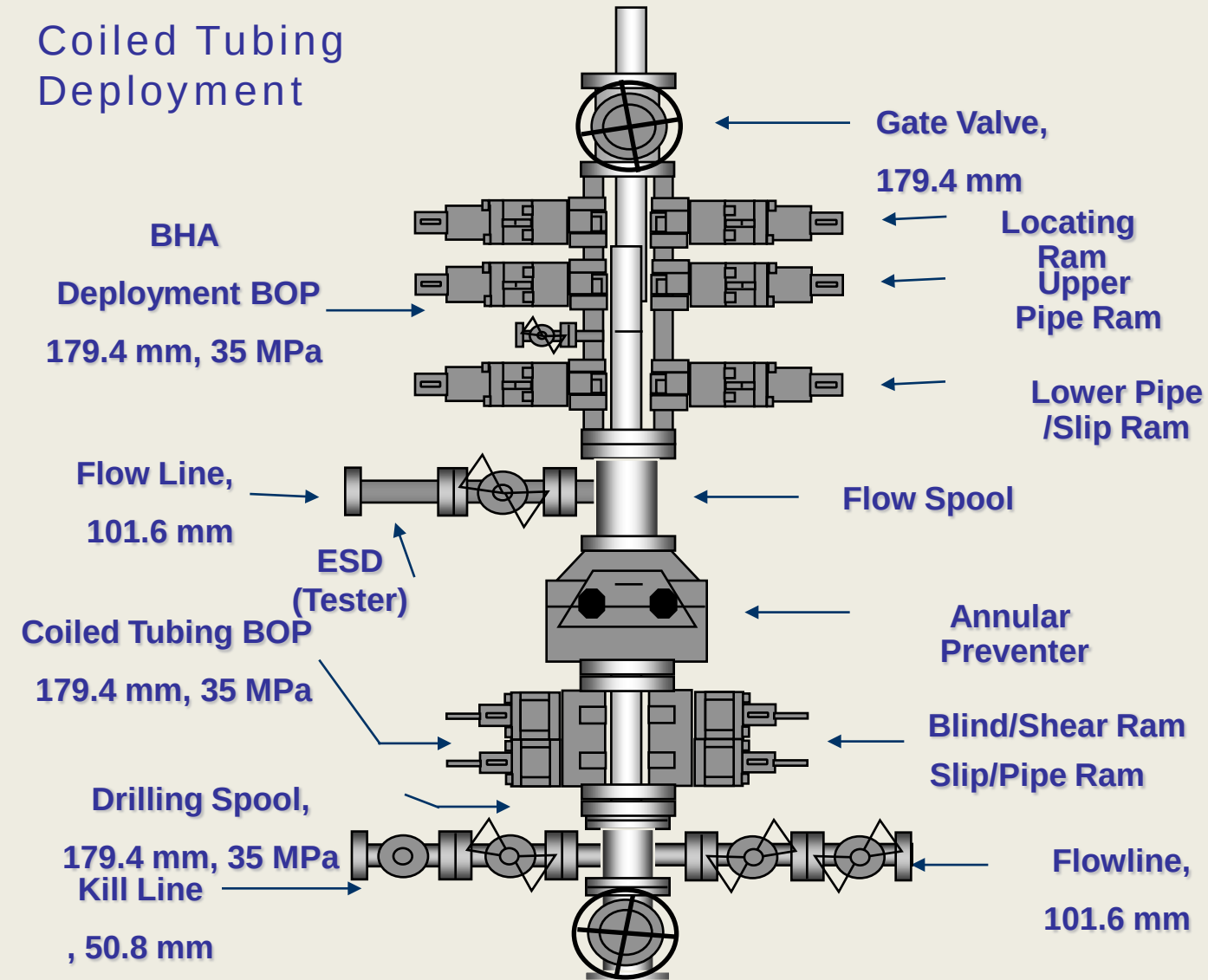
Blind/Shear Ram
Slip/Pipe Ram

Flowline,
101.6 mm

Deployment BOP

Conventional
Rig BOP

Wellhead

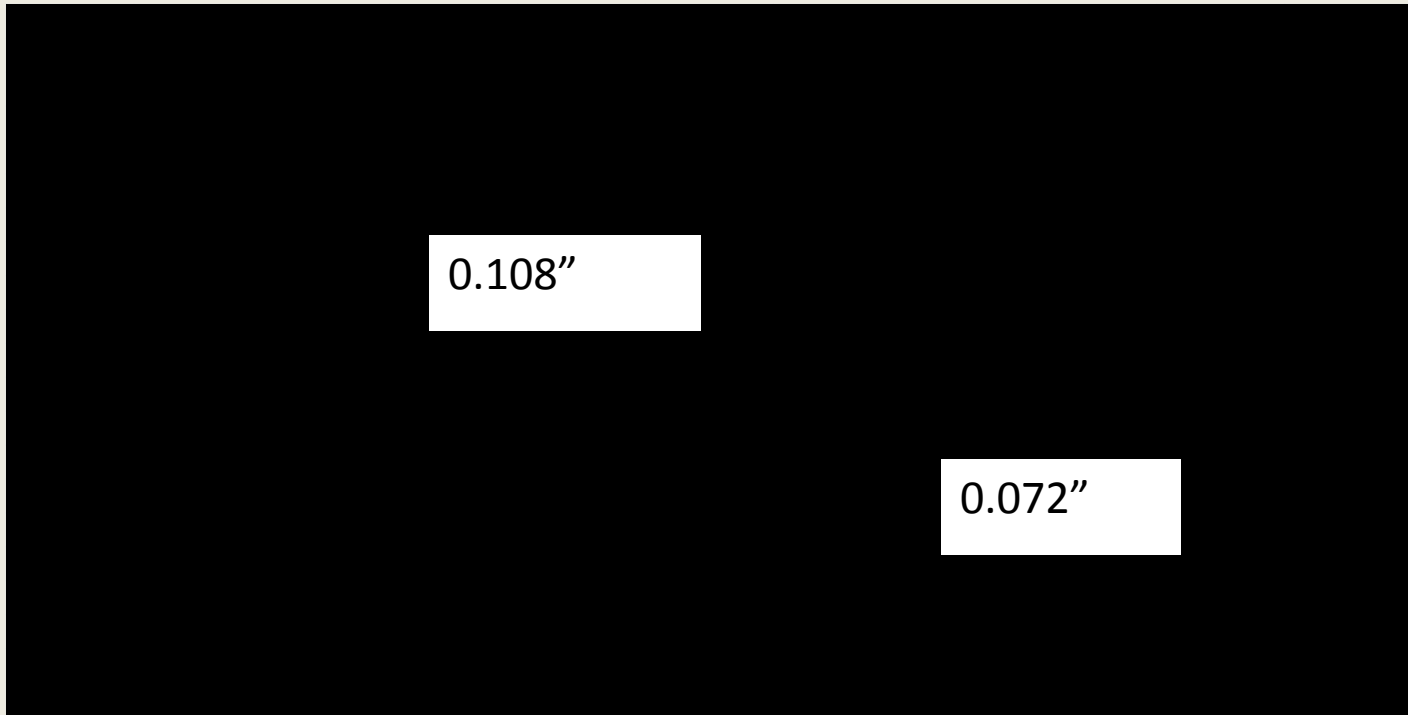


Physical Brace

- If there is a long space (length is a variable with pipe diameter, wall, and buckling force) then a brace between the bottom of the injector and the first seal is necessary.

Snubbing/Stripping Forces on CT

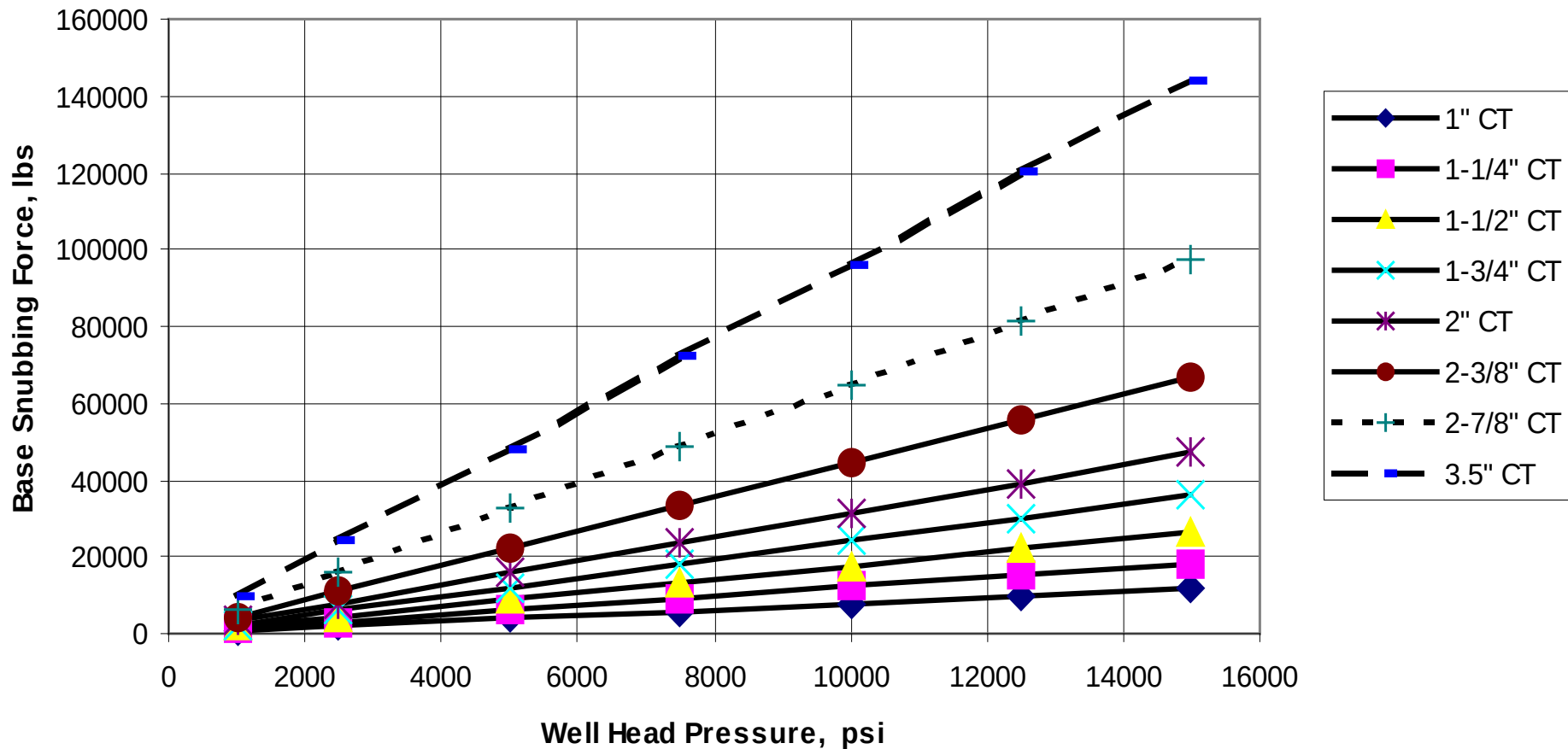
- Force to push CT through stuffing box/stripper (opposite running)
- Force on CT from Well Head Pressures - (upward)
- Force to overcome friction (opposite running)
- Force from weight of CT & BHA (downward)



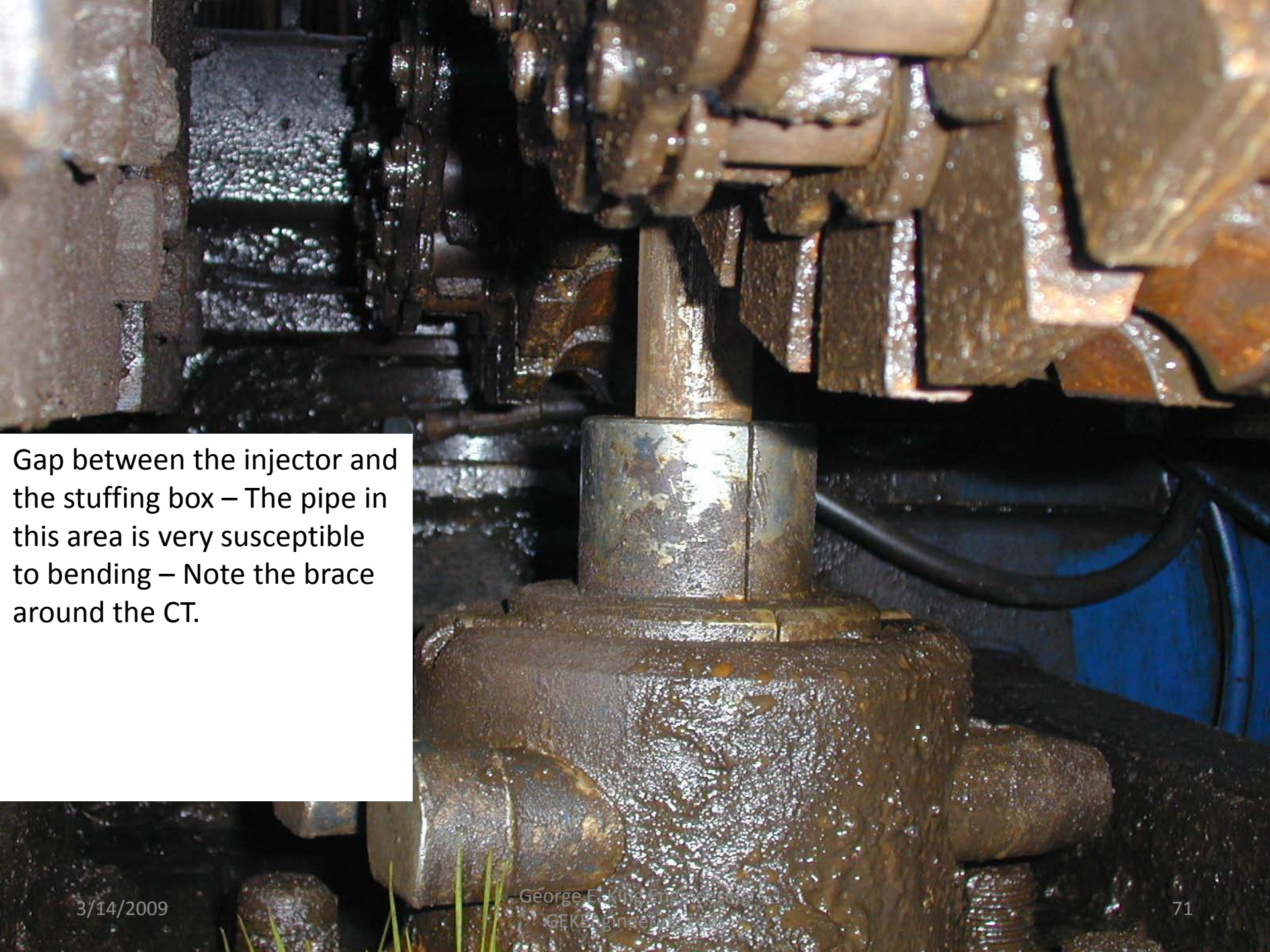
Problem in rig up height and lubricator length.

Snubbing Force on CT at Wellhead

Pressure Induced Snubbing Force VS Well Head Pressure for Various Coiled Tubing Sizes, (Must add stripper friction force)





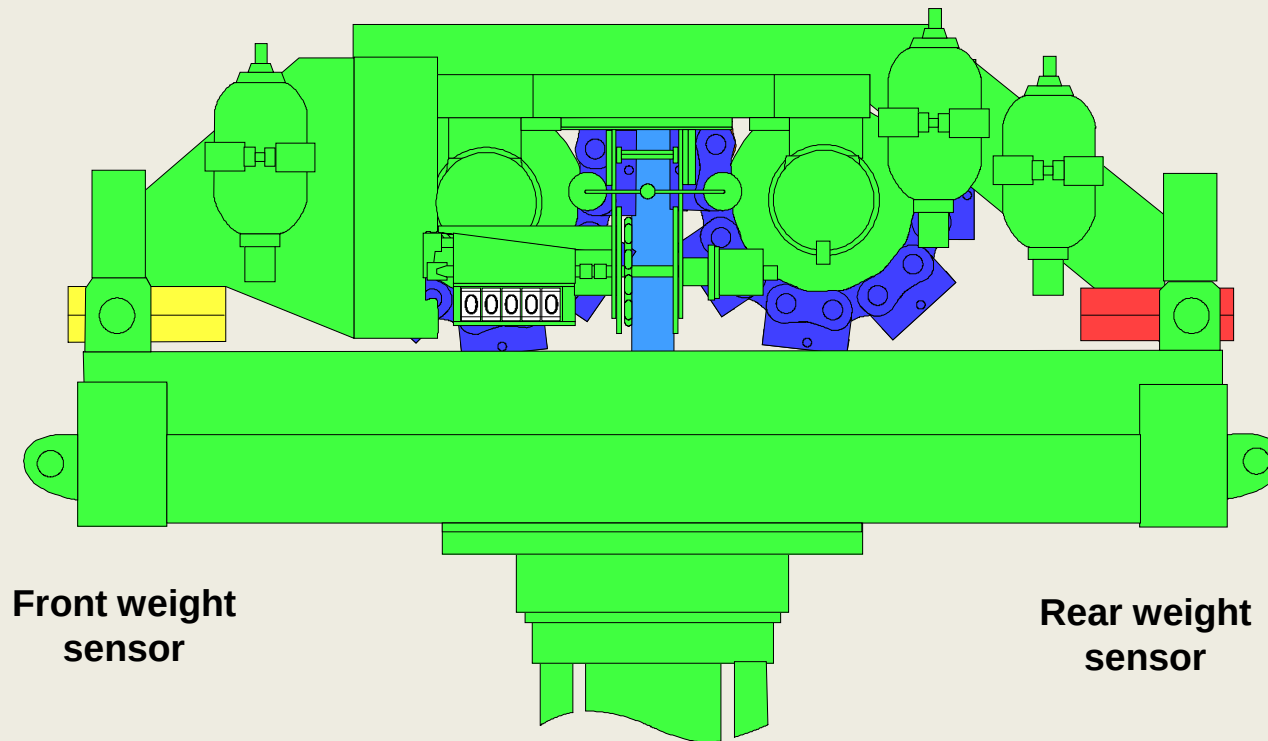


Gap between the injector and the stuffing box – The pipe in this area is very susceptible to bending – Note the brace around the CT.

Seals

- Stuffing Box - Primary sealing mechanism for isolating the wellbore. Positioned above the BOP, just underneath the injector head. A connector attaches the box to the BOP.
- BOP - multi-level control for closing in the well, shearing the CT, sealing around the CT and gripping the CT. May be used in pairs in extreme applications.

Weight Indicator Equipment



Typical Loads: 1.5" CT, 5000 psi

- snub force = $(1.767 \text{ in}^2 \times 5000 \text{ psi})$
=8835 lb
- stuffing box friction (drag force)
=2000 lb
- total
at start of the CT run-in =10835 lb

Load Changes w/CT inserted

- 1.5:1, 0.109" wall, 1.623 lb/ft
- one foot of CT in well = 10833 lb
- 1000' CT in well = $10835 - (1000 \times 1.623)$
= 9212 lbs
- 10,000' CT in well = $10835 - (10,000 \times 1.623)$
= -5395 lbs
- this assumes no wall friction